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In [2]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score
```

```
In [11]: df = pd.read_csv(r"D:\Pradnya\TE\DSBDA\Practicals\new\iris\iris.csv")

# Features and target
X = df.drop('Species', axis=1)
y = df['Species']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Train Naïve Bayes model
model = GaussianNB()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)
```

```
In [12]: # Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

# Function to compute TP, FP, TN, FN
def get_confusion_metrics(conf_matrix):
    FP = conf_matrix.sum(axis=0) - np.diag(conf_matrix)
    FN = conf_matrix.sum(axis=1) - np.diag(conf_matrix)
    TP = np.diag(conf_matrix)
    TN = conf_matrix.sum() - (FP + FN + TP)
    return TP, FP, TN, FN

tp, fp, tn, fn = get_confusion_metrics(cm)

# Performance metrics
accuracy = accuracy_score(y_test, y_pred)
error_rate = 1 - accuracy
precision = precision_score(y_test, y_pred, average='macro')
recall = recall_score(y_test, y_pred, average='macro')
```

```
In [13]: print("--- Confusion Matrix ---")
print(cm)

print("\nTrue Positives:", tp)
print("False Positives:", fp)
print("True Negatives:", tn)
print("False Negatives:", fn)

print("\n--- Performance Metrics ---")
```

```
print("Accuracy:", accuracy)
print("Error Rate:", error_rate)
print("Precision:", precision)
print("Recall:", recall)
```

--- Confusion Matrix ---

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

True Positives: [10 9 11]

False Positives: [0 0 0]

True Negatives: [20 21 19]

False Negatives: [0 0 0]

--- Performance Metrics ---

Accuracy: 1.0

Error Rate: 0.0

Precision: 1.0

Recall: 1.0

```
In [14]: # Plot confusion matrix
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=model.classes_,
            yticklabels=model.classes_)

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix Heatmap')
plt.show()
```

